

Engineering practice and automation

Slide 1 · 0-2 min

Engineering practice and automation

Every technique must discharge a named verification claim.

Make responsibilities
observable → Read machine-learning
evidence → Keep accountable
human gates

Teaching script

Open with a decision the audience must make in this area. Ask which result they currently treat as sufficient evidence and keep that answer for the closing discussion.

The lesson selects a practical argument from the book. The study handout preserves the detailed examples, citations and worked artifacts.

Source: ch24:13-34, ch25:25-72, ch25:247-311, ch26:13-38

Learning outcomes

Slide 2 · 2–5 min

Learning outcomes

Demonstrate these decisions on a real plan row.

- Assign ownership across verification handoffs.
- Interpret an ML result by task and denominator.
- Admit generated checkers through accountable review.
- Choose an improvement from the local verification bottleneck.

Teaching script

Read the outcomes as observable tasks. Ask learners to identify which one they can already demonstrate and which one currently depends on undocumented judgment.

At closure, require a concrete artifact or explanation for an outcome rather than a general expression of confidence.

Source: ch24:13–34, ch25:25–72, ch25:247–311, ch26:13–38

Route through the lesson

Slide 3 · 5–7 min

Route through the lesson

Four blocks, each ending with a decision.

- Make responsibilities observable
- Read machine-learning evidence
- Keep accountable human gates
- Choose the next improvement

Teaching script

Allow eighteen minutes for each block: a concept, its engineering consequence, a worked interpretation and a short discussion. Keep the discussion tied to the evidence available in the example.

Reserve the final eleven minutes for the assigned exercise and application to the learners' work. The timing is a course design allocation, not an engineering estimate.

Source: ch24:13–34, ch25:25–72, ch25:247–311, ch26:13–38

An artifact owner is not the whole outcome

Slide 4 · 7–11 min

An artifact owner is not the whole outcome

Integration seams need explicit responsibility.

- Name who resolves cross-team discrepancies.
- Keep the expected evidence visible at handoff.

Teaching script

Use a checker and model owned by different teams. Both can be delivered while their disagreement remains unresolved.

Assign responsibility for the outcome, including the interface between artifacts. The owner should be able to say which evidence permits the receiving team to proceed.

Source: ch24:13–34

Reviews answer different questions

Slide 5 · 11–16 min

Reviews answer different questions

Plan, code and closure reviews need different inputs.

Intent review → Implementation review → Closure review

Teaching script

Ask which review detects a missing requirement, which challenges a checker implementation and which accepts residual risk. Treating them as one generic approval loses those distinctions.

Bring the artifact that makes each question answerable. A coverage dashboard does not replace the specification at a plan review, and a code review does not authorize a waiver.

Source: ch24:35–86

Reproduction is an onboarding milestone

Slide 6 · 16–21 min

Reproduction is an onboarding milestone

A newcomer should establish evidence before explaining architecture.

- Start from a known failure and its reproducer.
- Produce a useful bug report and trace the ownership path.

Teaching script

A guided defect teaches the actual run environment, observations and team interfaces. Ask the newcomer to reproduce it without depending on an undocumented working directory.

Judge onboarding by a small useful artifact rather than vocabulary recall. The exercise also exposes missing process documentation in the team.

Source: ch24:87–104, ch24:105–186

Discussion: every deliverable has an owner

Slide 7 · 21–25 min

Discussion: every deliverable has an owner

Name the evidence that would settle the decision.

- The model and checker were both delivered.
- Their disagreement blocks integration.
- Which responsibility was left unnamed?

Teaching script

Expected answer: responsibility for resolving the combined verification outcome at the seam, including authority to settle intent.

Ask learners to define the evidence and escalation path that make the handoff complete. Merely adding another status meeting does not resolve ownership.

Source: ch24:13–34

Evaluate the proposed operational use

Slide 8 · 25–29 min

Evaluate the proposed operational use

A measured task outcome is more useful than generic accuracy.

- Use held-back data and a relevant comparison.
- Re-measure when the design or failure population changes.

Teaching script

Use the chapter's triage-classifier proposal. Agreement with the human triage decisions measures the task the classifier is being proposed to assist, within that held-back population.

A changed design or failure distribution can invalidate the estimate. Keep an owner, monitoring cadence and retraining cost with the deployment decision.

Source: ch25:25–72

A benchmark needs its denominator

Slide 9 · 29–34 min

A benchmark needs its denominator

Artifact class and acceptance criterion determine the meaning.

- Record filtering and excluded cases.
- Compare like tasks before comparing headline results.

Teaching script

The chapter compares studies producing different artifacts under different criteria. Ask what was counted as successful, what humans filtered, and what target set remained.

A result on one protocol or design does not establish a trend across workflows. Describe the reported experiment precisely before drawing a transfer claim for the local project.

Source: ch25:73–142

A generated checker can become a bad oracle

Slide 10 · 34–39 min

A generated checker can become a bad oracle

A proof against current RTL does not establish intended behavior.

- Inspect timing direction and activation.
- Challenge both design and generated interpretation.

Teaching script

A syntactically valid property may express the wrong temporal relation or never activate. A proof against the current design can confirm consistency while preserving the original intent error.

Ask what independent observation would reject the plausible but wrong property. Mechanical checking reduces the candidate set; it does not remove the need to inspect the intended claim.

Source: ch25:143–246

Discussion: most generated properties prove

Slide 11 · 39–43 min

Discussion: most generated properties prove

Name the evidence that would settle the decision.

- Generated properties pass against the current RTL.
- No one has checked them against intent.
- What has the reported result established?

Teaching script

Expected answer: consistency with that design under the proof environment, for the properties and candidates included in the measurement.

It has not established that the properties express the intended behavior or cover all required plan rows. Ask which acceptance work is still owed.

Source: ch25:73–142, ch25:143–246

Filter mechanically before intent review

Slide 12 · 43–47 min

Filter mechanically before intent review

Human attention should reach candidates with a clear provenance.

Mechanical checks → Intent review → Recorded admission

Teaching script

Use parsing, elaboration, signal resolution and available semantic checks to remove mechanical failures before asking a reviewer to assess intent.

The reviewer still decides whether the surviving property expresses the requirement and is admitted into the plan. A tool pass must not silently perform that admission.

Source: ch25:247–311

Correlated errors defeat casual sampling

Slide 13 · 47-52 min

Correlated errors defeat casual sampling

A shared generation mistake can affect an entire batch.

- Inspect systematic translation risks.
- Do not infer batch correctness from a few plausible examples.

Teaching script

Explain why repeated use of the same flawed template or interpretation creates dependent errors. A small informal sample does not establish the reliability of the unreviewed remainder.

Choose review and validation according to the artifact's role as an oracle. The chapter keeps acceptance authority with the people responsible for the verification claim.

Source: ch25:247-311

Preserve the accepted artifact, not just the model name

Slide 14 · 52-57 min

Preserve the accepted artifact, not just the model name

Provenance must support reconstruction of the decision.

- Retain relevant inputs, generated text and accepted revision.
- Record reviewer, rationale and applicable restrictions.

Teaching script

A model identifier does not identify the prompt context, retrieved specification, generation or human edits that produced the checker in use.

Keep the actual accepted artifact and the evidence supporting admission. Apply the project's data-handling rules to the inputs and outputs before using the workflow.

Source: ch25:312-438

Discussion: approval of a sampled batch

Slide 15 · 57-61 min

Discussion: approval of a sampled batch

Name the evidence that would settle the decision.

- A reviewer checks a few generated properties.
- All share the same translation template.
- What uncertainty remains about the batch?

Teaching script

Expected answer: a systematic error in the shared template or interpretation can affect many unreviewed properties. The sample's plausibility does not establish their intent correctness.

Ask for a validation action targeted at that shared failure mode and a traceable acceptance decision for the artifacts actually admitted.

Source: ch25:247-311, ch25:312-438

The bottleneck depends on the project level

Slide 16 · 61–65 min

The bottleneck depends on the project level

A block and a software workload can need different improvements.

- Separate knowing what to check from execution capacity.
- Measure the local constraint before investing.

Teaching script

Contrast a block regression that already fits overnight with a system workload that cannot reach its useful software state in time. More execution capacity addresses only one of those immediate constraints.

The chapter's claim is conditional on level and workload. Do not turn it into a universal prediction about every verification team.

Source: ch26:13–38

Earlier and later evidence are complementary

Slide 17 · 65–70 min

Earlier and later evidence are complementary

Moving a check does not erase its original scope obligations.

- Use early models with explicit fidelity limits.
- Retain deep block checking alongside system execution.

Teaching script

Earlier models expose integration or software problems before RTL is ready; later platforms exercise behaviors that need long execution or physical observations.

Neither movement justifies dropping a plan row without replacement evidence for the same claim. Review the handoff of model assumptions and residual behavior.

Source: ch26:39–54

Automation moves work without removing judgment

Slide 18 · 70–75 min

Automation moves work without removing judgment

Partial oracles and model boundaries remain engineering problems.

- Improve the weakest evidence-producing step.
- Keep interpretation, review and ownership explicit.

Teaching script

Separate contingent tool capacity from durable limitations such as incomplete specifications and partial observation. Better automation can change the cost of a task without deciding what correctness should mean.

Ask learners to name a proposed improvement and the claim it strengthens. This keeps future-looking discussion tied to an inspectable engineering outcome.

Source: ch26:55–85, ch26:86–174

Discussion: cut regression after a new proof

Slide 19 · 75–79 min

Discussion: cut regression after a new proof

Name the evidence that would settle the decision.

- A new formal result motivates removing simulation work.
- Its model covers only part of the old claim.
- What evidence is required before the cut?

Teaching script

Expected answer: show that the retained and replacement evidence jointly discharge the original plan row, including behavior outside the proof model.

A proof's strength does not make it transferable to an unrelated boundary. Preserve the residual simulation obligation or explicitly revise the claim with a reviewed risk decision.

Source: ch26:39–54

Exercise: compare two automation proposals

Slide 20 · 79–85 min

Exercise: compare two automation proposals

Distinguish task evidence from design consistency.

- One classifier is measured against held-back human triage.
- Generated properties instead prove against current RTL.
- Explain which result measures its proposed use and what remains unknown.

Teaching script

Exercise 25.2 (judgement). The chapter opens with two proposals reaching one verification lead in the same week. The first offers a failure classifier for the video codec's nightly regression, which agreed with the human triage engineer 88% of the time on a held-back month of history. The second offers a specification-to-assertion flow piloted on the crypto engine, most of whose generated properties passed a formal proof against the current RTL. Decide which of the two reports evidence about the use it proposes, and give the property of the first measurement that the second one lacks. State what the passing proofs do establish, pointing to the section that explains why a proof verdict settles consistency rather than intent.

Assign book Exercise 25.2 in full. Use the chapter's stated held-back comparison without treating its illustrative result as a prediction for another project.

Expected answer: the classifier measurement evaluates the proposed operational task against independent human triage on held-back history. Passing proofs show consistency with the modeled RTL and assumptions; they do not establish that generated properties express the intended specification.

Source: ch25:25–72, ch25:143–246

What to change on Monday

Slide 21 · 85–87 min

What to change on Monday

Make one existing verification decision reviewable.

- Repair one unowned integration handoff.
- Audit provenance for an accepted generated checker.
- Measure the bottleneck before selecting an automation project.

Teaching script

Ask each learner to choose one of these actions in an existing project. Require a small, inspectable deliverable and identify the person who will review it.

Keep the change connected to the lesson's evidence limits. A new dashboard, script or model is useful only if it strengthens the claim the plan row is meant to establish.

Source: ch24:13–34, ch25:25–72, ch25:247–311, ch26:13–38

Scope and next reading

Slide 22 · 87–90 min

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Engineering practice and automation handout and book Chapters 24–26.
- Model training recipes and predictions about future product performance are outside this lesson.

Teaching script

Revisit the outcomes and ask the room to state a limitation of the strongest result discussed today. A complete answer identifies a model boundary, an assumption, a missing oracle or the scope of the underlying evidence.

Assign the detailed reading to the relevant chapters. The omitted topics are explicit scope boundaries, not claims that those topics are unimportant.

Source: ch24:13–34, ch25:25–72, ch25:247–311, ch26:13–38